

N100 Finance Intelligence Platform

Analyst Guide

Version: 1.0

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Project:

Bluestock Fintech Internship

Technology Stack:

- Python
- SQLite
- FastAPI
- Pandas
- NumPy
- Pytest
- ReportLab

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The N100 Finance Intelligence Platform is a financial analytics platform developed as part of the Bluestock Fintech Internship. The project provides a complete pipeline for collecting, processing, analyzing, and serving financial data for Nifty 100 companies.

The platform is designed to help investors, analysts, and researchers quickly access financial information through a REST API and automatically generated reports.

Major objectives of the project include:

- Build an ETL pipeline to process financial datasets.
- Store structured financial data in SQLite.
- Calculate financial KPIs for every company.
- Expose analytics through FastAPI endpoints.
- Generate professional PDF company tearsheets.
- Ensure data quality using automated validation.
- Maintain code quality using unit testing and linting.

Technology Stack

- Python 3.12
- SQLite Database
- FastAPI
- Pandas
- NumPy
- ReportLab
- Pytest
- Black
- Ruff

Project Architecture

Excel Files



ETL Pipeline



SQLite Database



Financial KPI Engine



FastAPI



Reports / API Consumers

This project is developed using Python and SQLite. Follow the steps below to set up the project on a local machine.

Prerequisites

- Python 3.12 or later
- Git
- SQLite
- Visual Studio Code (Recommended)

Project Setup

1. Clone the repository

```
git clone https://github.com/<your-username>/N100-Finance-Intelligence-Platform.git
```

2. Navigate to the project folder

```
cd N100-Finance-Intelligence-Platform
```

3. Create a virtual environment

```
python -m venv .venv
```

4. Activate the virtual environment

Windows:

```
.venv\Scripts\activate
```

5. Install project dependencies

```
pip install -r requirements.txt
```

Running the ETL Pipeline

The ETL (Extract, Transform, Load) pipeline reads financial Excel datasets, validates the data, and loads it into the SQLite database.

To execute the ETL pipeline:

```
python src/etl/database_loader.py
```

Expected Outcome

- Financial datasets are loaded successfully.
- SQLite database is updated.
- Validation checks are completed.
- The database is ready for API queries.

Database Structure

The N100 Finance Intelligence Platform uses SQLite as its primary database for storing financial and market data. The database is populated through the ETL pipeline and serves as the backend for API requests and analytics.

Core Tables

- companies

Stores company master information such as company ID, name, sector, and other metadata.

- sectors

Contains sector classifications for companies.

- profitandloss

Stores revenue, expenses, operating profit, net profit, and earnings information.

- balancesheet

Contains assets, liabilities, shareholder equity, borrowings, and reserves.

- cashflow

Includes operating, investing, and financing cash flow data.

- financial_ratios

Stores calculated financial KPIs generated by the analytics engine.

- stock_prices

Historical stock price information used for analysis and reporting.

Financial KPI Engine

The KPI Engine computes key financial metrics for each company using the stored financial statements.

Implemented KPIs include:

- Return on Equity (ROE)
- Return on Capital Employed (ROCE)
- Return on Assets (ROA)
- Net Profit Margin
- Operating Profit Margin
- Debt-to-Equity Ratio
- Interest Coverage Ratio
- Asset Turnover Ratio
- Free Cash Flow
- Revenue CAGR
- PAT CAGR
- EPS CAGR
- Composite Quality Score

These KPIs enable analysts to evaluate profitability, efficiency, leverage, growth, and overall financial health across companies.

The N100 Finance Intelligence Platform exposes financial data through a FastAPI REST API. The API allows analysts and developers to retrieve company information, financial metrics, sector summaries, and screening results.

Starting the API Server

Run the following command from the project root:

```
uvicorn src.api.main:app --reload
```

The API starts on:

<http://127.0.0.1:8000>

Interactive API Documentation

Swagger UI:

<http://127.0.0.1:8000/docs>

ReDoc:

<http://127.0.0.1:8000/redoc>

Example Endpoints

1. Health Check

GET `/api/v1/health`

Example:

```
curl http://127.0.0.1:8000/api/v1/health
```

2. Company Profile

GET /api/v1/companies/TCS

Example:

```
curl http://127.0.0.1:8000/api/v1/companies/TCS
```

3. Stock Screener

GET /api/v1/screener?min_roe=15

Example:

```
curl "http://127.0.0.1:8000/api/v1/screener?min_roe=15"
```

4. Sector Summary

GET /api/v1/sectors

5. Companies by Sector

GET /api/v1/sectors/Information%20Technology/companies

API Response Format

The API returns responses in JSON format, making it easy to integrate with dashboards, web applications, and other analytics tools.

The N100 Finance Intelligence Platform can generate professional PDF reports (tearsheets) for individual companies as well as batch reports for multiple companies.

Single Company Tearsheet

Run the following command:

```
python src/reports/tearsheet.py
```

This generates a PDF report containing:

- Company Overview
- Financial Performance
- Profitability Metrics
- Growth Indicators
- Leverage Ratios
- Cash Flow Analysis
- Composite Quality Score

Batch Tearsheet Generation

To generate reports for all available companies:

```
python src/reports/batch_tearsheet.py
```

The generated PDF files are saved in the output directory for further analysis or sharing.

Benefits

- Standardized company reports
- Easy offline viewing

- Professional presentation format
- Useful for investors and analysts

The project includes automated tests to ensure correctness and reliability.

Testing Frameworks

- Pytest
- FastAPI TestClient
- Unit Tests
- Integration Tests

Running the Test Suite

`pytest tests/`

Code Formatting

`black src tests`

Static Code Analysis

`ruff check src tests`

Continuous testing helps identify issues early and ensures the application remains stable as new features are added.

During development and deployment, users may occasionally encounter issues. The following table lists common problems and their solutions.

1. Database Connection Error

Problem:

SQLite database cannot be found.

Possible Cause:

- Incorrect database path.
- Database file missing.

Solution:

- Verify the database location.
- Check the database path in the configuration file.
- Ensure the ETL pipeline has been executed successfully.

2. Import Errors

Problem:

ModuleNotFoundError or ImportError.

Solution:

- Activate the virtual environment.
- Install all required packages using:

```
pip install -r requirements.txt
```

3. API Not Starting

Problem:

FastAPI server fails to launch.

Solution:

- Verify port 8000 is available.
- Close any process already using the port.
- Restart the server:

```
uvicorn src.api.main:app --reload
```

4. Test Failures

Problem:

Some Pytest cases fail.

Solution:

- Review the error message.
- Ensure the database has the latest data.
- Re-run:

```
pytest tests/
```

5. PDF Generation Errors

Problem:

Tearsheet generation fails.

Solution:

- Verify ReportLab is installed.
- Check output directory permissions.
- Ensure company data exists in the database.

Best Practices

- Keep dependencies updated.
- Run tests before committing changes.
- Use Black and Ruff before every release.
- Backup the SQLite database regularly.

The current version of the N100 Finance Intelligence Platform provides a strong foundation for financial analytics. Future versions can include additional features to improve usability, scalability, and analytical capabilities.

Planned Enhancements

1. Interactive Dashboard

Develop a web-based dashboard for visualizing financial KPIs, sector comparisons, and company performance.

2. Advanced Stock Screening

Support multiple screening criteria, including valuation ratios, momentum indicators, dividend yield, and custom filters.

3. Portfolio Analytics

Allow users to create and monitor investment portfolios with performance tracking and risk analysis.

4. Real-Time Market Data

Integrate live market data APIs to provide up-to-date stock prices and financial metrics.

5. Machine Learning Models

Implement predictive analytics for forecasting financial performance and identifying investment opportunities.

6. Cloud Deployment

Deploy the application using cloud services with CI/CD pipelines for automated testing and deployment.

These enhancements will make the platform more useful for analysts, investors, and researchers.

The N100 Finance Intelligence Platform demonstrates the complete development of a financial analytics solution, from data ingestion to API delivery and automated reporting.

Throughout this project, the following objectives were achieved:

- Built an ETL pipeline for financial datasets.
- Designed and populated a SQLite database.
- Calculated financial KPIs for Nifty 100 companies.
- Developed FastAPI endpoints for data access.
- Generated PDF company tearsheets.
- Implemented automated testing for reliability.
- Improved code quality through formatting and linting.
- Documented the project for analysts and developers.

The platform provides a scalable foundation for future enhancements and can be extended with additional analytics, dashboards, and deployment capabilities.

Thank you for using the N100 Finance Intelligence Platform.